Lab 4: Dynamics of Voltage-Gated Ion Channels

Stream 1: Cell Signalling

August 28, 2026

1 Goal

To model the dynamics of voltage-gated ion channels using ordinary differential equations (ODEs). You will implement models for potassium (activation only) and sodium (activation and inactivation) channels, and explore through simulation why the separation of timescales and the voltage-alignment of gating variables are critical for generating a transient sodium current.

2 Reading material

Keener, J., & Sneyd, J. (2025). *Mathematical Physiology* (3rd ed.). Springer.

- Chapter 3: Membrane Ion Channels, particularly Section 3.5 (Channel gating)

3 Guided example: a simple model of the K^+ channel

You will build a model for the potassium channel conductance (g_K) based on the Hodgkin-Huxley formalism, where the channel has four identical and independent activation subunits.

Your Python script `lab4_channels.py` should be structured to match the steps below. The gating dynamics of a K^+ channel are described by:

$$\tau_n(V)\frac{dn}{dt} = n_\infty(V) - n \tag{1}$$

$$g_K(t) = \bar{g}_K n(t)^4 \tag{2}$$

where $n(t)$ is the probability of a single subunit being open, and $g_K(t)$ is the total channel conductance. We will use simplified sigmoidal functions for the voltage-dependent steady-state (n_∞) and time constant (τ_n):

$$n_\infty(V) = \frac{1}{1 + e^{-(V-V_0)/10}}$$

$$\tau_n(V) = 1 + 5e^{-(V-V_0)^2/50^2}$$

Use the following physiological parameters: $\bar{g}_K = 1.0$ (mS/cm^2), $V_0 = -25.0$ (mV),

Part A: solving the ODE for the gating variable

First, simulate how the activation variable n evolves during a voltage-clamp experiment.

1. **Define the System in Python:** Create a function for the ODE (1) that describes dn/dt. It should take time (t), the state variable (n), and the membrane voltage (V) as arguments.
2. **Solve numerically:** Use the `compute` function to integrate the system starting at a steady state. Simulate a voltage step from a holding potential of $V = -70$ mV to a test potential of $V = 20$ mV at $t = 5$ ms.
   - First, find the steady-state value of n at $V = -70$ mV to use as your initial condition $n(0)$.
   - Simulate for a total of 50 ms.
3. **Plot the results:** Generate a time series plot showing how $n(t)$ evolves and approaches its new steady state after the voltage step.

Part B: calculating the total conductance

Next, use your solution for $n(t)$ to find the conductance of the entire channel.

1. **Calculate conductance:** Using your array of $n(t)$ values from Part A, calculate $g_K(t)$ at each time point using equation (2).
2. **Plot the conductance:** Create a plot of the potassium conductance, $g_K(t)$, over time. Notice its sigmoidal (S-shaped) rise. This shape is fundamentally different from the simple exponential rise of $n(t)$ that you plotted in Part A.

4 Exercise: dynamics of the Na^+ channel

Next, you will model the sodium channel to understand the critical role of voltage alignment. The sodium channel conductance is described by an activation (m) and an inactivation (h) gate:

$$g_{Na}(t) = \bar{g}_{Na} m(t)^3 h(t)\,.$$

The gating variables are governed by their own independent ODEs:

$$\tau_m(V)\frac{dm}{dt} = m_\infty(V) - m \tag{3}$$

$$\tau_h(V)\frac{dh}{dt} = h_\infty(V) - h\,, \tag{4}$$

where we choose steady-state functions

$$m_\infty(V) = \frac{1}{1 + e^{-(V - V_0^m)/k_m}}$$

$$h_\infty(V) = \frac{1}{1 + e^{(V - V_0^h)/k_h}}$$

and parameters $\bar{g}_{Na} = 1.0$ (mS/cm^2), $V_0^m = -30.0$ (mV), $V_0^h = -50.0$ (mV), $k_m = 8$, and $k_h = 5$. Define the time constants as: $\tau_m(V) = 0.5$ ms and $\tau_h(V) = 5.0$ ms. Note that $\tau_m \ll \tau_h$.

Your task is to adapt your Python script to simulate the sodium channel and demonstrate why a functional Na^+ channel also requires the voltage ranges of activation and inactivation to be correctly aligned to create a 'window current'. You will run two experiments.

Experiment 1: The realistic case

1. **Modify the code:**
   - Implement a function that solves the two independent ODEs for m and h simultaneously.
2. **Generate plots and analyse:**
   - Simulate the system for a voltage step from $V = -70$ mV to $V = 20$ mV. Remember to use the correct steady-state values at -70 mV as your initial conditions.
   - On a single figure, plot the gating variables $m(t)$ and $h(t)$ versus time, and on a second subplot, the total conductance $g_{Na}(t)$ versus time.
   - Observe the transient 'window' where the conductance rises and then falls.
3. **Visualise the window current:**
   - Plot the steady-state activation ($m_\infty(V)$) and inactivation ($h_\infty(V)$) curves on the same graph over a voltage range of -100 mV to 40 mV.
   - On a second subplot, plot the product $m_\infty(V)^3 \cdot h_\infty(V)$. This product is proportional to the steady-state 'window current'. Identify the voltage range where this window exists.

Experiment 2: The importance of the voltage window

1. **Create a hypothetical 'misaligned' channel:**

   (a) modify the inactivation steady-state function to be:

   $$h_\infty(V) = \frac{1}{1 + e^{(V - V_0^h)/k_h}}$$

   with $V_0^h = -70$ (mV).

   (b) modify the activation steady-state function to be:

   $$m_\infty(V) = \frac{1}{1 + e^{-(V - V_0^m)/k_m}}$$

   with $k_m = 0.5$.

2. **Generate plots and analyse:** For each case (a) and (b), re-run the full simulation from Experiment 1 but with this misaligned $h_\infty(V)$ respectively $m_\infty(V)$.

   - Re-create the plots from experiment 1 using this new misaligned $h_\infty(V)$ respectively $m_\infty(V)$ function. What has happened to the window current?
   - Plot the resulting conductance $g_{Na}(t)$.

5 'Advanced': singular perturbation

The large difference between τ_m and τ_h allows us to use a powerful mathematical technique called singular perturbation analysis to simplify the model. Because the activation variable m is so much faster than h, we can approximate it as being instantaneously at its steady-state value, $m_\infty(V)$. This is known as the quasi-steady-state approximation (QSSA).

Task: test the QSSA

1. **Define the reduced model:** The QSSA reduces the model for the sodium conductance to:

   $$g_{Na,\text{approx}}(t) = \bar{g}_{Na} \cdot [m_\infty(V(t))]^3 \cdot h(t)$$

   Note that we still need to solve the ODE for the slow variable, $h(t)$.

2. **Modify your Python script:** Take your code from Experiment 1 (the realistic case).

   - After solving for $h(t)$, calculate the approximated conductance, $g_{Na,\text{approx}}(t)$, at each time point using the formula above. For the voltage step, $V(t)$ is simply the constant test potential of 20 mV.

3. **Compare and analyse:**

   - On a single graph, plot both the full conductance from Experiment 1, $g_{Na}(t)$, and the approximated conductance, $g_{Na,\text{approx}}(t)$.

6 Interpretation tasks

Discuss the outcomes with your peers and interpret your simulation results. Use the plots generated by your script to answer the following questions.

1. **K^+ channel dynamics:** Compare the plot of the activation variable $n(t)$ with the plot of the total conductance $g_K(t)$. Why does the conductance rise with a sigmoidal (S-shape) curve, while the underlying gating variable $n(t)$ rises exponentially? What is the biophysical interpretation of the n^4 term?

2. **Na^+ channel timescales:** In Experiment 1 for the Na^+ channel, you observed a transient conductance. Explain the roles of the fast activation gate (m) and the slow inactivation gate (h) in producing this shape. What would happen to the sodium current if τ_m and τ_h were equal?

3. **Na^+ channel voltage alignment:** In Experiment 2, you investigated two hypothetical misaligned channels.

   - **Case (a):** When you shifted the inactivation curve (h_∞) to more negative voltages ($V_0^h = -70$ mV), the transient conductance disappeared. Explain why, with reference to your plot of the steady-state window current.
   - **Case (b):** When you made the activation curve (m_∞) steeper ($k_m = 2$), the transient conductance was largely unaffected, even though the steady-state window current appeared to be closed. Explain this apparent contradiction by considering the time constants (τ_m and τ_h). Why is the separation of timescales just as important as the voltage alignment?